Measurement Environment for the EMC Characterization of Power Semiconductor Devices

Von der Fakultät für Informatik, Elektrotechnik und Informationstechnik der Universität Stuttgart zur Erlangung der Würde eines Doktor-Ingenieurs (Dr.-Ing.) genehmigte Abhandlung

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Declaration

I, Robert Kragl, hereby declare that this submission – except for the supervision by the Institute for Power Generation and Storage Systems (PGS) of RWTH Aachen University – is solely my own work. Information derived from the published or unpublished work of others has been acknowledged in the text and a list of references is given.

Aachen, April 10, 2026

Contents

1 Introduction

The design of power electronic systems inherently involves trade-offs among several competing objectives: efficiency, compactness, reliability, cost, and electromagnetic compatibility (EMC) [24, 27].

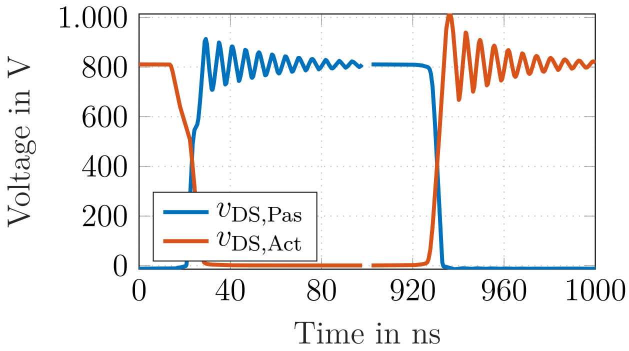
Among these, EMC estimation often presents the greatest challenge during development due to its strong dependence on a wide range of system- and component-level parameters. As a result, EMC is frequently underemphasized during early development. Consequently, nominally optimized converter designs are often reworked late in the development cycle—typically by reducing switching speed or restructuring the layout of the commutation loop. This reactive mitigation degrades efficiency and power density, increases development time and cost, and shifts effort from systematic design to ad hoc redesign. Early-stage characterization of EMC performance at both component and system level is therefore required.

In EMC analysis, it is well established that the switched voltage of power semiconductor devices is the primary source of both conducted and radiated emissions [19, 22]. However, how this noise is propagates within in the system depends on the passives and the layout [21]. Therefore, EMC analysis can be divided into two components: (1) noise generation by the semiconductor and (2) noise propagation through the passive circuit elements, including the heatsink, to the measurement ports – typically a Line Impedance Stabilization Network (LISN) for conducted or an antenna for radiated emissions [8, 18, 28].

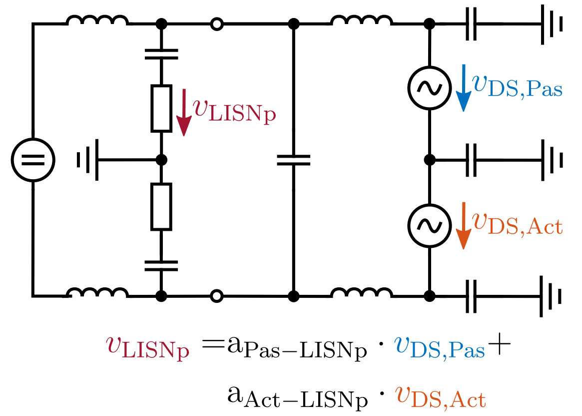
In respsect to the noise propagation a lot of research has be conducted to mititgate the propagation within the system. For example by optimizing the passive layout of the commutation loop [5], through filters [8, 9], the integration of additional ground capacitance [4, 30] or shielding layers [10, 34].

In respect to the noise source, two different frequency ranges must be considered differently. For frequencies below 10 MHz, the switched voltage can be modeled using idealized trapezoidal waveforms [3, 19] (see Fig. 1), which provide adequate accuracy for EMC evaluation in this frequency range. Engineers can improve EMC performance by modifying converter operation – such as adjusting switching frequency or the modulation strategy [30].

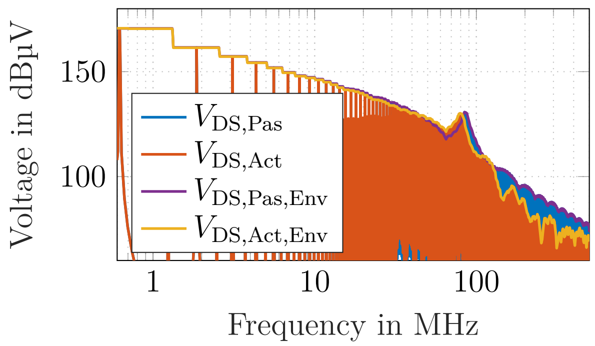
However, at frequencies above 10 MHz, EMC analysis becomes significantly more complex. In this regime, the intrinsic switching characteristics of the semiconductor devices dominate the emission spectrum, making accurate modeling and prediction more difficult [12]. This dependence on switching behavior introduces challenges – such as production-induced variability and operating-point sensitivity – while also creating opportunities for targeted electromagnetic noise optimization in future device and system developments.



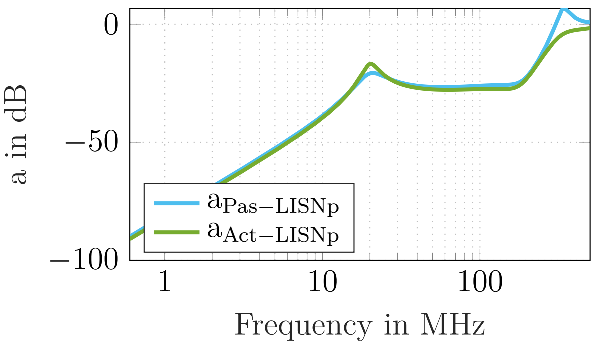
(a) Active and passive switching signals in time domain.



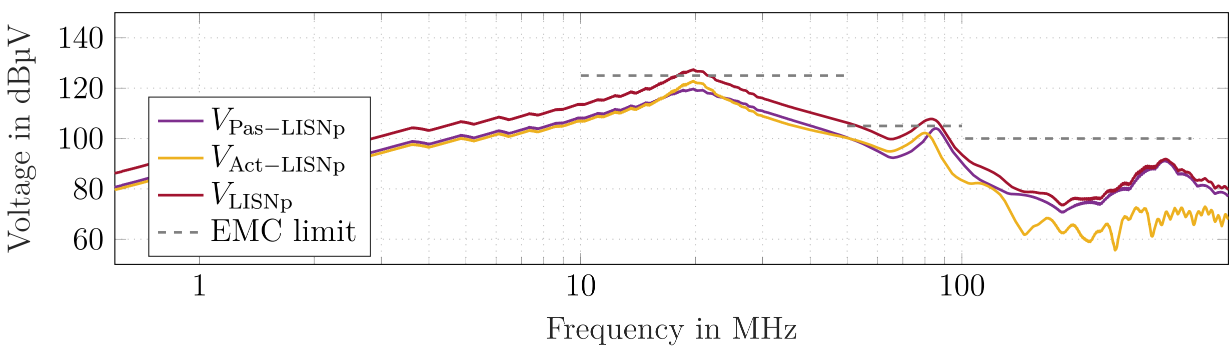
(b) Simplified circuit diagram of half-bridge converter.



(c) Spectrum of active and passive switch.



(d) Artificial transfer function from passive and active switch to LISN measurement port.



(e) Effective LISN spectrum generated by passive and active switch.

Figure 1.1: Schematic explanation of noise propagation within a half-bridge converter. Simulated switching waveforms and artificial transfer functions.

As a result, extensive research has focused on characterizing and modeling these high-frequency effects. Broadly, this research can be divided into two methodological approaches: system-level experimental measurements and model-based simulations.

In experimental setups, various semiconductor technologies are tested within a commutation cell, and their emissions are measured using a LISN [1, 10, 16, 28, 29, 33] or antenna [5, 23]. This enables broad semiconductor technology comparisons. However, these measurements only capture the conglomerated noise from all noise sources of the system, making it impossible to distinguish between individual switches or even specific switching events within the same device. Moreover, the results are highly sensitive to the passive setup surrounding the semiconductor, which influences how noise propagates to the LISN or antenna.

To gain deeper insight into the influence of semiconductor behavior on EMC performance, some researchers have turned to simulation-based approaches. These studies vary behavioral models to identify parameters that affect emissions in specific frequency bands [7, 13, 17, 32]. However, these models are typically parameterized using data from classical double-pulse tests and static measurements [2, 13], which do not capture the high-frequency behavior of switches beyond their resonance frequency. As this work will demonstrate, such limitations render the results of these simulations unverifiable beyond the device resonance frequency.

This represents a fundamental challenge in current semiconductor-focused EMC analysis: existing experimental methods cannot isolate individual device contributions, while simulation approaches lack the high-frequency fidelity required for verification. A measurement methodology is needed that can directly characterize the intrinsic electromagnetic emissions at the semiconductor power terminals with sufficient bandwidth and particularly sufficient signal-to-noise ratio.

Standard drain-source voltage measurements with passive or differential probes become noise-limited above the device resonance frequency (typically 10 MHz to 100 MHz depending on chip size [12, 15]), while EMC issues in practice can occur up to 400 MHz to 500 MHz. This upper limit reflects practical EMC laboratory experience rather than established scientific criteria. Above this range, observed EMC issues are typically driven by digital clocking (e.g., microcontrollers, FPGAs) rather than by power semiconductor switching.

An initial approach for measuring device noise spectra above the resonance frequency was introduced in [20]. This method employed a passive probe combined with a high-pass filter to isolate high-frequency components of the switching event. Building on this concept, [11] extends the setup to simultaneously measure both high-side and low-side chips using differential probes.

This contribution picks up the a methodology, which enables the separated EMC analysis of all switching events, presented in [12], and applies it to the voltage sensor presented in [11]. Additionally, a current sensory is added, which is suitable for EMC measurements. To the authors' knowledge, this is the first demonstration of simultaneous intrinsic semiconductor emission measurement at the device terminals together with concurrent system-level EMC acquisition at a LISN.

This measurement approach addresses three critical limitations of existing methods: (1) it isolates the individual noise contribution of each device, (2) it extends the observable frequency range beyond device resonance, and (3) it maintains correlation with conventional EMC metrics through integrated LISN measurements.

Through direct characterization of the intrinsic semiconductor emissions, this setup enables meaningful comparisons between different semiconductor technologies in terms of their EMC behavior. Therefore, it provides valuable insights early in the development process, offering opportunities to improve the EMC performance of future semiconductor generations through design parameters such as doping profiles, metallization, and trench architecture.

This paper is organized as follows: Section II presents the methods needed for the measurement of EMC-relevant noise emissions directly at the semiconductor ports. Section III introduces measurement setup with particular focus on the verification of all new measurement probes. Section IV demonstrates the approach through comparative measurements of different semiconductor technologies, while Section V concludes the work.

2 Fundamentals

2.1 EMC in Power in Electronics

2.2 Dynamic Measurements for Power Electronics

Loading

Small Signal Response

Large Signal Response

Parasitic Coupling

Transmission Lines

3 Methods for Measurement of EMC-Relevant Signal Components

The objective is to measure intrinsic electromagnetic noise emissions of power semiconductor devices directly at their power terminals. As will be seen this measurement requires additional measurement methods, since measurements with classical approaches can not capture the relevant frequency range, since the signal falls below the noise floor.

3.1 Measurement Limitation

Commercially available voltage probes cannot measure the EMC-relevant frequency band of 10 MHz to 400 MHz of SiC MOSFETs. This limitation is not due to the bandwidth or resolution constraints of the probes, but rather due to the nature of the spectrum of switched voltages and the subsequent sampling process. For explanation, Fig. 3.1 illustrates a measurement of a drain-source voltage and its frequency spectrum.

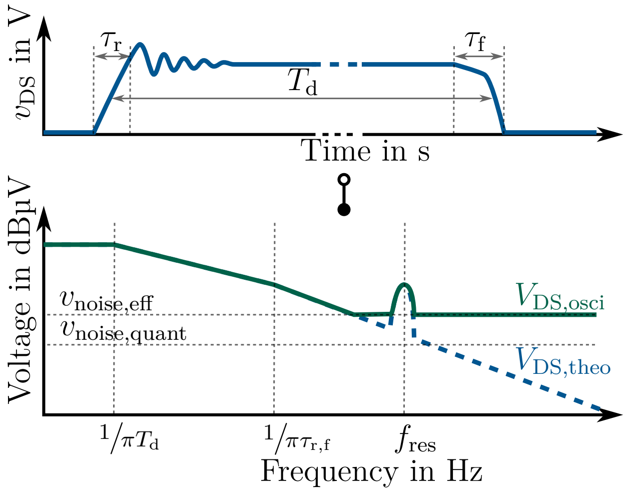


Figure 3.1: Limitation of conventional voltage measurement and its impact on the frequency spectrum.

In the frequency range below $1/\pi T_{\mathrm{d}}$ the spectra corresponds to the DC component of the signal $V_{\mathrm{DS}}(f = 0) = \mathrm{d} \cdot V_{\mathrm{DC}}$. Above $f_{\mathrm{cut1}} = 1/\pi T_{\mathrm{d}}$, the spectrum declines with -20 dB/dec. This region is influenced by the switching frequency, modulation scheme and operating point of the converter. At frequencies above $f_{\mathrm{cut2}} = 1/\pi\tau_{\mathrm{r}}$, another transition occurs in the spectra, with the ideal voltage spectra falls with -40 dB/dec. This is the region, where the switching transient and, thereby, the semiconductor is the dominating factor for the spectrum behavior. Above $f_{\mathrm{cut2}} = {}^{1}/_{\pi t_{\mathrm{r}}}$ the switching transient and, thereby, the semiconductor is the dominating factor for the spectrum.

The challenge in the measurement of this frequency band, can be seen in Fig. 3.1. Above the resonance frequency, the signal drops below the noise floor, making it impossible to derive any informative value regarding the emitted noise. An exemplary calculation can be done with the spectra formulation from graphical analysis of a symmetric trapezoidal pulse [3].

$$V_{\mathrm{DS}}(f) = \mathrm{d}V_{\mathrm{DC}} \cdot \mathrm{sinc}(\pi f \mathrm{d}T_{\mathrm{sw}}) \cdot \mathrm{sinc}(\pi f \tau_{\mathrm{r}}) \tag{3.1}$$

$$\leq \begin{cases} \mathrm{d}V_{\mathrm{DC}} & , \quad 0\,\mathrm{Hz} \leq f < \dfrac{1}{\pi \mathrm{d}T_{\mathrm{sw}}} \\ \dfrac{V_{\mathrm{DC}}}{\pi f T_{\mathrm{sw}}} & , \quad \dfrac{1}{\pi \mathrm{d}T_{\mathrm{sw}}} \leq f < \dfrac{1}{\pi \tau_{\mathrm{r}}} \\ \dfrac{V_{\mathrm{DC}}}{\pi^2 f^2 T_{\mathrm{sw}} \tau_{\mathrm{r}}} & , \quad \dfrac{1}{\pi \tau_{\mathrm{r}}} \leq f \end{cases} \tag{3.2}$$

Now, lets assume some value as: $V_{\mathrm{DC}} = 800\,\mathrm{V}$, $\mathrm{d} = 0.5$, $f_{\mathrm{sw}} = 25\,\mathrm{kHz}$, $\tau_{\mathrm{r}} = 20\,\mathrm{ns}$ and $\tau_{\mathrm{f}} = 50\,\mathrm{ns}$. As a result

$$V_{\mathrm{DS}}(f = 100\,\mathrm{MHz} = 4000\ f_{\mathrm{sw}}) = 14.2\,\mathrm{mV} \tag{3.3}$$

A classical probe with an attenuation of 250 is used, which results in an ADC input voltage

$$v_{\mathrm{ADC,in}} = \frac{v_{\mathrm{meas}}}{\mathrm{a}_{\mathrm{probe}}} = \frac{14.18\,\mathrm{mV}}{250} = 57\,\mu\mathrm{V} \tag{3.4}$$

The problem is evident when the minimum voltage increment of the oscilloscopes ADC is calculated, with an ADC, with an effective number of bits (ENOB) of 14

$$v_{\mathrm{ADC,min}} = \frac{V_{\max} - V_{\min}}{\mathrm{a}_{\mathrm{probe}} \cdot 2^{\mathrm{N_{bit}}}} = \frac{1200\,\mathrm{V} - (-150\,\mathrm{V})}{250 \cdot 2^{14}} = 330\,\mu\mathrm{V}. \tag{3.5}$$

The two values are of the same order of magnitude, which makes it for the ADC impossible to sample this voltage components. Additional to the quantization noise some thermal noise adds to the noise floor, creating a higher, effective noise floor, as can be seen in Fig. 3.1.

$$v_{\mathrm{noise,eff}} = v_{\mathrm{noise,quant}} + v_{\mathrm{noise,therm}} \tag{3.6}$$

add johnson noise formula

The start of the noise floor is normally a little above the resonance frequency of output capacitance of the MOSFET and commutation loop inductance, which is around 10 MHz to 50 MHz for modules and 100 MHz for discretes and, therey, much lower than the bandwidth of conventional probes. Hence, the limitation is not in the resolution or bandwidth of the probes, but in the sampling process of the spectrum, see Fig. 3.1.

At this point it must be highlighted that these voltage components are only of interest for EMC considerations. For classical considerations, as overvoltage, the components are so small that they do not have any impact.

3.2 Oversampling and Averaging

The previous section explains the challenges to measure high-frequency components of a signal. Firstly, classical digital noise reduction can be applied, which address the thermal noise $v_{\mathrm{noise,therm}}$: Oversampling and averaging.

By oversampling and averaging afterwards, the effective number of bits (ENOB) can be increased by [14].

$$\mathrm{N_{ENOB}} = \mathrm{N_{ADC}} + 0.5 \cdot \log_2\left(\frac{f_\mathrm{s}}{2 f_\mathrm{max}}\right) \tag{3.7}$$

Here N_{ADC} is the nominal ADC resolution, f_s is the sampling frequency of the oscilloscope and f_max the maximum frequency of interest. Furthermore, the factor of 2 in the denumerator is due to the nyquist sampling theorem. This can be done internally in the oscilloscope.

As this cannot be pushed limitless due to hardware-restrictions of the maximum sampling speed of modern oscilloscopes, the number of signal points can be increased by not measuring one single pulse, but a pulse chain of M pulses. This further increases the ENOB to [11]

$$\mathrm{N_{ENOB}} = \mathrm{N_{ADC}} + 0.5 \cdot \log_2\left(\frac{f_\mathrm{s}}{2 f_\mathrm{max}}\right) + 0.5 \log_2(M) \tag{3.8}$$

These gains are realized only if the dominant noise is random, uncorrelated with the signal, and greater than the quantization step [**silicon_laboraties_an118_2013**]. As discussed above, in power electronics the high-frequency voltage components often fall below or near the ADC quantization level; therefore oversampling and averaging alone are insufficient and must be combined with a high-pass measurement method introduced next.

3.3 High-Pass Measurement

To overcome this measurement limitation and measure the EMC-relevant voltage components directly at the semiconductors ports a high-pass measurement can be executed as visualized in Fig. 3.2. The high pass is designed to rise with +40dB/dec to counteract the decreasing slope of the drain-source voltage (-40dB/dec). As a result, the input voltage at the ADC has a flat frequency spectrum, which makes it perfect for the sample process of the ADC.

Diese formeln sehen noch nicht ganz korrekt aus...

$$v_{\mathrm{DS,HP,raw}} = a_{\mathrm{HP}} \cdot v_{\mathrm{DS}} \tag{3.9}$$

To calculate the real spectrum of the drain-source voltage, the measured signal has to be compensated with the transfer function of the probe. This is normally done in the frequency domain since here the compensation is a simple addition.

$$V_{\mathrm{DS,HP,comp}} = \mathcal{F}\{a_{\mathrm{HP}} \cdot v_{\mathrm{DS}}\} - A_{\mathrm{HP}} \tag{3.10}$$

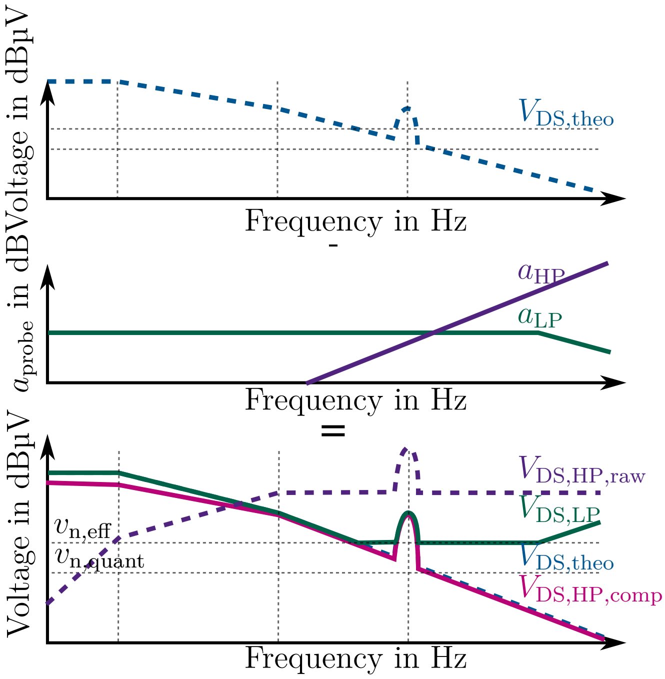
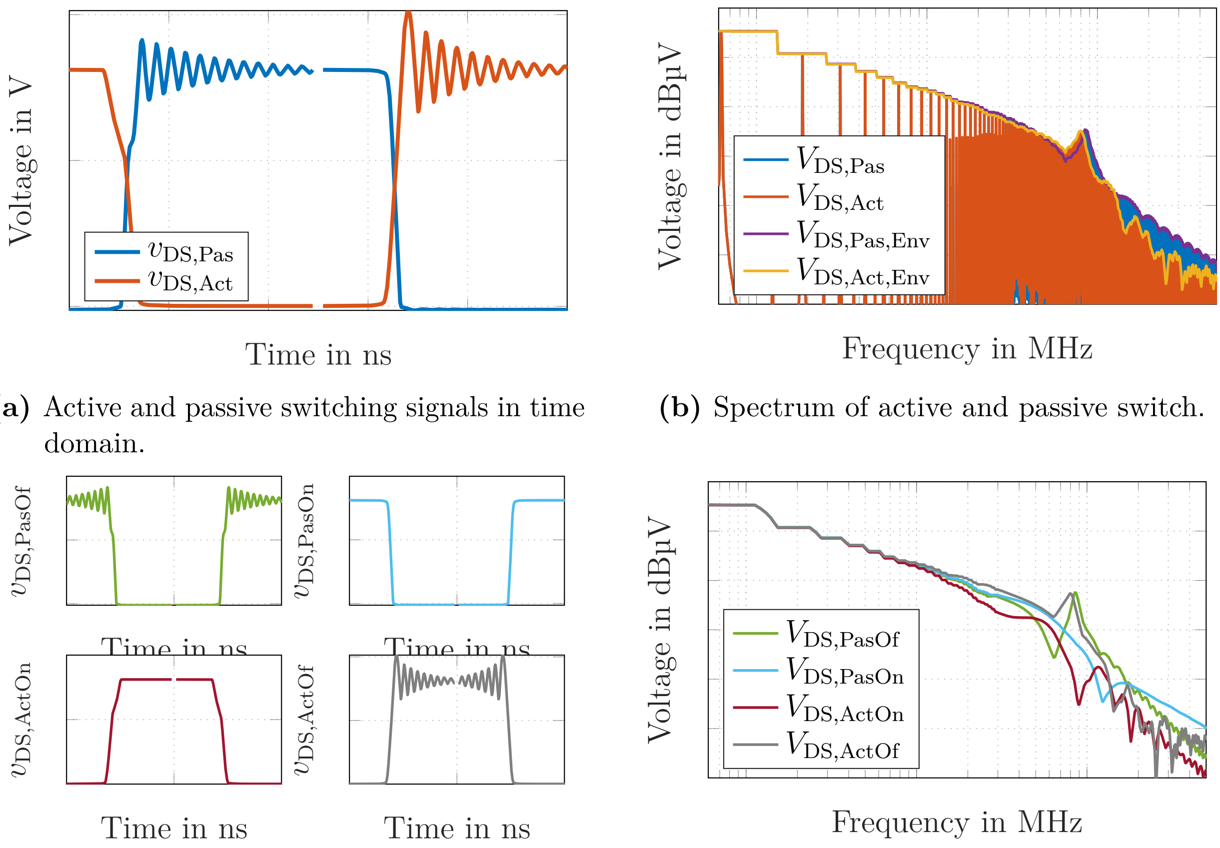


Figure 3.2: Limitation of conventional voltage measurement and its impact on the frequency spectrum.

As a result, the measurement is no longer limited by the start of the noise floor, but only the bandwidth of the used probe.

3.4 Switching Event Separation

In the analysis of the switching behavior of power electronics it is common practice to analyze the four switching events separately - active-off/passive-on and active-on/passive-off [6]. Here, the active device is the switch, which initiates the switching transient while the passive switch is turned off and the body diode operates during the commutation process. This separated analysis is also desirable for the EMC analysis to identify the switching event



(a) Active and passive switching signals in time domain.

(b) Spectrum of active and passive switch.

(c) Snapped and mirrored time domain signals.

(d) Spectrum of snapped and mirrored signals.

that is dominating the frequency spectrum for the respective frequency band of interest. This is enabled by-and one of the main advantages-of the measurement directly at the semiconductor ports. Conventional EMC measurements at LISNs or antennas can only measure the conglomerated noise of all switching events and noise sources in the system.

However, to achieve this separated analysis in the frequency domain, a method introduced in [12] must be used, called snap-and-mirror function.

This function identifies the mid-points of each switching cycle $t_{\mathrm{mid,n}}$ and mirrors them around this axis, which can be formulated as

$$\mathrm{s_{new}}(t) = \begin{cases} s(t) & \text{, for } t_{\mathrm{mid,n}} < t < t_{\mathrm{mid,n+1}} \\ s(-t + 2 \cdot t_{\mathrm{mid,n+1}}) & \text{, for } t_{\mathrm{mid,n+1}} < t < t_{\mathrm{mid,n+2}} \\ s(t) & \text{, for } t_{\mathrm{mid,n+2}} < t < t_{\mathrm{mid,n+3}} \\ s(-t + 2 \cdot t_{\mathrm{mid,n+3}}) & \text{, for } t_{\mathrm{mid,n+3}} < t < t_{\mathrm{mid,n+4}} \\ \dots \end{cases} \tag{3.11}$$

This is done for both the turn-on and turn-off event for both the active and the passive

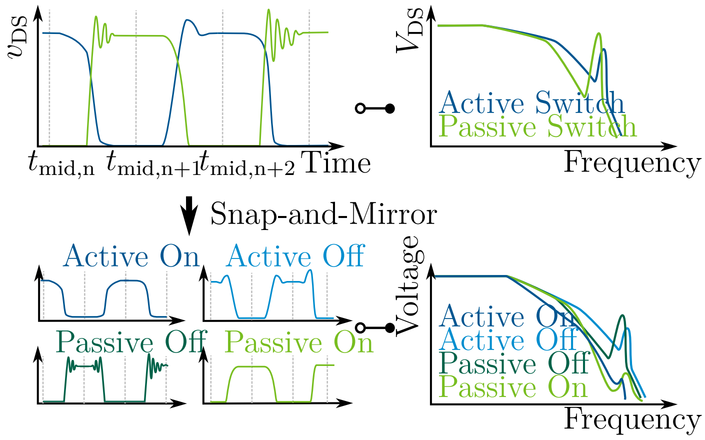


Figure 3.4: Principle functionality of snap and mirror function for the separated EMC analysis of all four switching events.

switch, thereby creating four artificial signals out of the two measured voltages of active and passive switch.

Because only the signal amplitude in the frequency domain is analyzed this is a valid operation since the mirroring and the time shift of a signal has only influence on the phase of the fourier transform.

$$\mathcal{F}\{s(-t)\} = S(-f) = S^*(f) \tag{3.12}$$

$$\mathcal{F}\{s(t-t_0)\} = S(f)e^{-j\omega t_0} \tag{3.13}$$

4 Dual-Channel Differential Active Voltage Probe

4.1 Requirements

4.2 Design

4.2.1 Frequency compensated voltage divider

4.2.2 Amplification

For the measurement of the drain-source voltage, a dual-channel differential active voltage probe is utilized. This probe enables simultaneous measurement of the drain-source voltages across both switching devices. The probe is connected directly to the respective drain and source terminals, without the use of any cables, which change their impedance and transfer function in dependence of their positioning. Furthermore this approach minimizes parasitic effects.

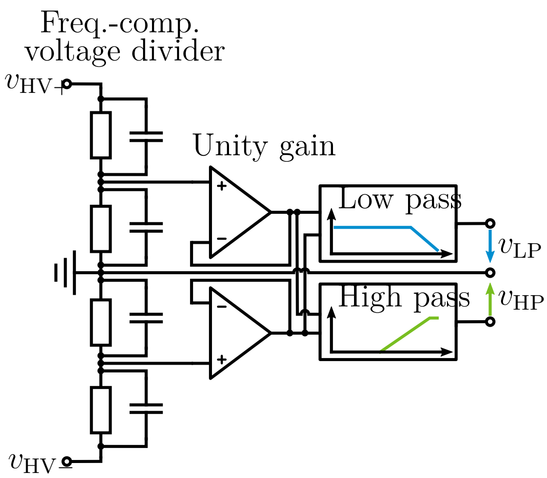


Figure 4.1: Schematic overview of probe functionality.

In contrast to conventional differential probes the signal is split after the unity gain amplifier to a second channel, which has a 2nd order high-pass characteristic. The probe is a new version of the design of [11] incorporating an additional shielding layer in a now 6-layer PCB. Furthermore, the HV PCB capacitor can be now trimmed by adding or removing the effective area of the outer capacitor side. The new design was verified with the methods introduced in [11] for both large signal as small signal response. The VNA measurement results are additionally shown in Fig. 4.2 anf Fig. 4.3.

As was explained in [11] each probe is verified with 3 measurements: VNA Small signal, Pulse Generator Large signal, and i-situ measuremnt in parallel with commercial measurement (for lowpass measurements - 80 mR passive shunt and 10 MR passive probe). the last is necessary for the verification of not having any parasitic capacitive or inductive couplings into the measurement path.

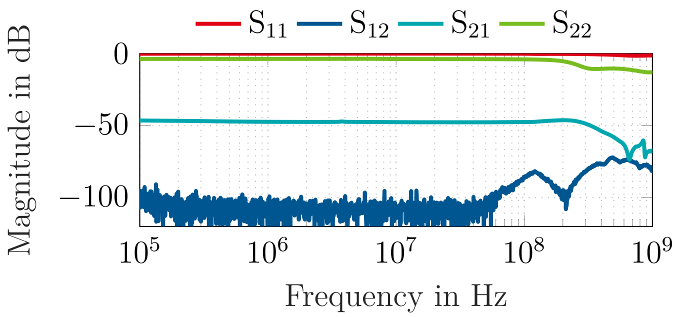


Figure 4.2: VNA measurement of low pass.

The voltage gain a_{probe} of the probe can be calculated from the S-parameter measurement as

$$a_{probe} = \frac{v_{out}}{v_{HV+} - v_{HV-}} \tag{4.1}$$

$$= \frac{S_{21}}{1 + S_{11}}. \tag{4.2}$$

Since S_{11} remains mostly flat at 0 dB due to the high-impedance input of the probe, the primary parameter of interest for the voltage gain is S_{21}.

The low-pass signal exhibits a bandwidth of approximately 250 MHz before the gain begins to drop, a limitation attributed to the characteristics of the operational amplifier employed in the output stage.

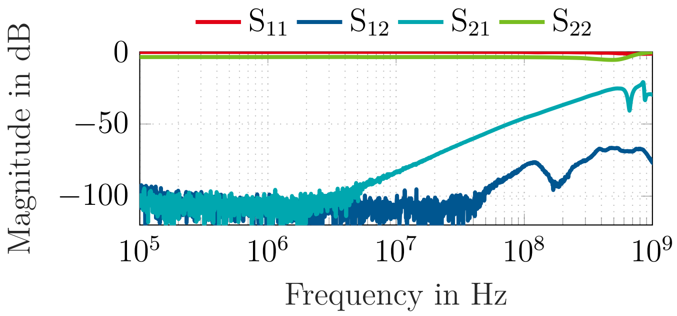


Figure 4.3: VNA measurement of high pass.

Considering the high pass transfer function the constant rise of gain over frequency was achieved with an additional input inductance in front of the unity gain amplifier. This compensates capacitive input behavior of the used

4.3 Verification

5 Active Current Shunt

While for lower frequencies mainly the capacitive coupling through the switched potential of the phase potential to the cooler dominate EMC concerns, at higher frequencies also inductive couplings occur. Therefore it is necessary to measure not only the high-frequency components of the voltage transient but also of the current transient.

5.1 Requirements

For the current measurement some requirements must be defined for a measurement in respect to EMC. First-same as for the voltage measurement- a bandwidth of 500 MHz or higher is needed. This makes the use of commercial Rogowski-coils impossible, furthermore rogoskis have the disadvantage to being prone to the positioning of the coil [31].

Furthermore, a galvanic isolated measurement is necessary, since the ground of the oscilloscope is fixed to the aluminum plate, which is also the ground of the LISNs. Hence, classical passives shunts, which set the oscilloscope ground to DC+ or DC- are not feasible. Lastly, the introduced measurement resistance must be one decade smaller than the $R_{\mathrm{DS,on}}$ of the DUT. The presented measurement setup is designed for 750 V and 1200 V SiC-discretes in a D2PAK where the $R_{\mathrm{DS,on}}$ 20 mW to 50 mW. (Cite PMTW05 and PMTX03 datasheets). Therefore, the sense resistance must be 2 mΩ or smaller. This is a problem for conventional shunts when the bandwidth is considered. Every shunt resistance has some parasitic inductance and the bandwidth of the shunt corresponds to

$$f_{\mathrm{bw,shunt}} = \frac{1}{2\pi\tau_{\mathrm{shunt}}} = \frac{R_{\mathrm{shunt}}}{2\pi L_{\mathrm{shunt}}} \tag{5.1}$$

Therefore, in classical shunt design higher bandwidths can be achieved with higher shunt resistance values. However, this is not feasible for the EMC measurement.

5.2 Design

Therefore, a replica of an active shunt of [25, 26] was desgined, which equivalent circuit is shown in Fig. 5.1. This provides multiple advantages: First a small shunt value can be used, while still maintaing a rather high sense output due to the amplification of the operational amplifier. Furthermore, the low pass filter of the active shunt compensates the high pass characteristic of the shunt resistance achieving remarkable bandwidth. Lastly, the active shunts creates a 50 W signal, which can be either sent to a filter stage for high pass EMC measurements or be measured with a optical-isolated voltage probe and thereby providing galvanic isolation.

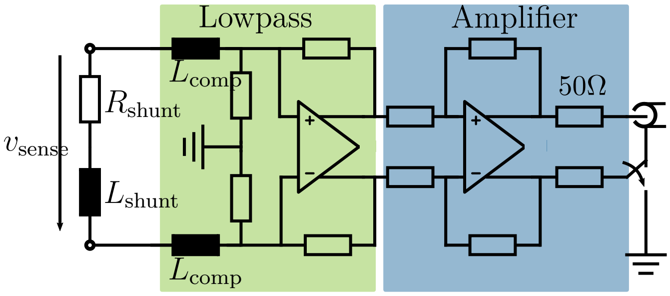


Figure 5.1: Schematic overview of active current shunt.

S21 » S12, um passive parasitäre übertragung zu negieren.

5.2.1 Shunt

5.2.2 Amplification

5.3 Verification

6 Test Bench

The measurement setup aims to incorporate the methods discussed in the previous section to measure the EMC-relevant signal components of drain-source voltages and the drain current. Furthermore a conventional EMC measurement through two LISNs is installed to compare the measurements results of the intrinsic EM emissions of the switch with conventional measurements.

A picture of the measurement setup is shown in Fig. 6.1 and its equivalent circuit is depicted in Fig. 6.2.

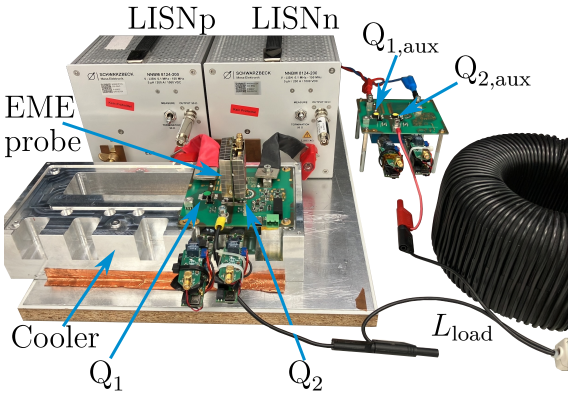


Figure 6.1: Photograph of measurement setup. (Change to new picture including two probes and the active shunt)

6.1 Requirements

6.2 Design

The two DUTs Q_1 and Q_2 are placed on a cooler, which is mounted on a aluminum plate, which creates the current path for the ground currents to the two LISNs.

As will be seen later, a second half-bridge with $Q_{1,aux}$ and $Q_{2,aux}$ is placed adjacent to the DUTs. This half-bridge is needed, because a single double pulse test is not sufficient for EMC measurements. Instead multiple pulses which are averaged in post-processing are needed to decrease the thermal noise. However, the EM noise of the auxiliary half-bridge should not be seen on the LISN measurements and, hence, it is not placed on the aluminum plate. The DC ports of the auxiliary half-bridge are additionally isolated from the DUTs through the impedance of the LISNs and of the inductor.

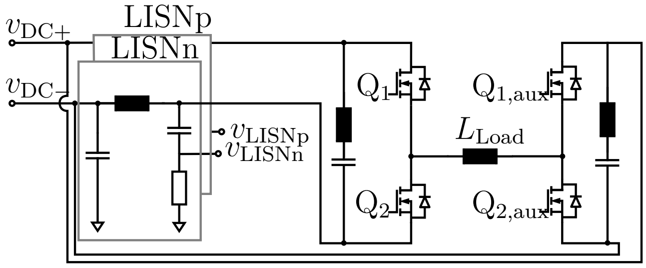


Figure 6.2: Equivalent circuit diagram of the back-to-back converter measurement setup. Add vds and id lp hp signals

6.2.1 Back-to-Back Converter

6.2.2 LISN Measurement

7 Limitations of Measurement Environment

7.1 Common-Mode Rejection Ratio (CMRR)

7.2 Crosstalk

7.3 Signal Roll-Off

8 Measurement Results

8.1 Chip Size Area

8.2 Technology Comparison

8.3 Passives

9 Conclusion

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